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Energy Conversion

Assignment 4

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1 Problem 1

a)

Motional emf:

$$E = BLv = 0$$

$$I = \frac{V_B - E}{R} = \frac{V_B}{R} = 314.3 \text{ A}$$

$$\rightarrow F = BLI = 283.9 \text{ N}$$

b)

$$I = 0$$

$$\rightarrow V_B = E = BLv \rightarrow v = 122.2 \text{ m/s}$$

c)

$$F = 30 \text{ N}$$

$$\rightarrow BLI = 30 \text{ N} \rightarrow I = 33.33 \text{ A}$$

$$E = V_B - IR = 98.33 \text{ V}$$

$$\rightarrow v = 109.3 \text{ m/s}$$

Mechanical power:

$$P_{out} = F \ell$$

Electrical power:

$$P_{in} = V_B I$$

$$\rightarrow \eta = \frac{P_{out}}{P_{in}} = 89.44\%$$

2 Problem 2

a)

$$BL\bar{I} = 11 \text{ N} \rightarrow \bar{I} = 52.38 \text{ A}$$

$$E = V_B - IR = 96.19 \text{ V} = BL\mathcal{U}$$

$$\mathcal{U} = 458 \text{ m/s}$$

b)

$$I = \frac{F_a}{BL}$$

$$E = V_B - IR \rightarrow \mathcal{U} = \frac{V_B - \frac{F_a}{BL} R}{BL}$$

$$\mathcal{U} = 507.70 \text{ m/s}$$

c)

$$\mathcal{U} = 285.5 \text{ m/s}$$

d) { Magnetic density
Voltage source

3 Problem 3

a)

$$A = (2r)l$$

$$\begin{aligned} e(t) &= BA\omega \sin(\omega t) \\ &= 16.46 \sin(105t) \text{ V} \end{aligned}$$

b)

$$i(t) = \frac{e(t)}{R} = 1.372 \sin(105t) \text{ A}$$

$$I_{\text{rms}} = 0.970 \text{ A}$$

c)

$$\tau(t) = BA i(t) \sin \omega t$$

$$= 0.215 \sin^2(105t) \text{ N.m}$$

It opposes the mechanical rotation.

4 Problem 4

In an AC machine, the electromagnetic torque always acts to reduce the angle γ . The stator field is lagging behind the rotor field. The induced torque acts in the opposite direction of the rotor's rotation. The torque trying to decelerate the rotor. The machine acts as a generator.

5 Problem 5

a)

$$n_s = \frac{120f}{p}$$

Poles	Speed (rpm)
2	1500
4	750
6	500
8	375
10	300
12	250
14	214.3

b)

$$P_n \propto f$$

$$P_e \propto f^2$$

$$\frac{P_{n,25}}{P_{n,60}} = 41.7\%$$

$$\frac{P_{e,25}}{P_{e,60}} = 17.4\%$$

c)

Light flicker decreases as frequency increases.

At 25 Hz, Light flicker is noticeable.

d)

$$P_{elec} = I_{rms}^2 R = 11.3 \text{ W}$$

e)

$$P_{mech} = T_{avg} \omega = \frac{T_{max}}{2} \omega = 11.3 \text{ W}$$

P_{elec} and P_{mech} are equal as expected from conservation of energy